

Designing a Feature Extraction Method Based on Causality Graph



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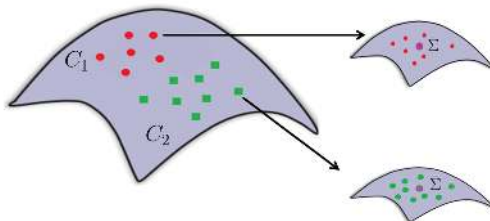


Summary



- Extraction submatrices of EEG signals, using a timevarying sliding window.
- Computing sample covariance matrix for i^{th} window ($\tilde{\Sigma}_i$).
- Computing the average sample covariance matrix using Riemannian metric on S.P.D. manifold:

$$\bar{\Sigma} = \arg \min_{\Sigma \in \mathcal{M}_{\text{spd}}} \sum_i d_{\mathcal{R}}^2(\Sigma, \tilde{\Sigma}_i) \quad (1)$$





- $\tilde{\Sigma}_i$ are **not full-rank!**
- We use Riemannian metric.

$$d_{\mathcal{R}}(\tilde{\Sigma}_1, \tilde{\Sigma}_2) = \left\| \ln(\tilde{\Sigma}_1^{-1} \tilde{\Sigma}_2) \right\|_F = \sqrt{\sum_n \ln^2(\lambda_n)} \quad (2)$$

Where λ_n is n^{th} eigenvalue of $\tilde{\Sigma}_1^{-1} \tilde{\Sigma}_2$.

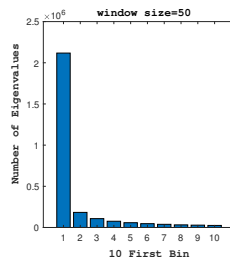
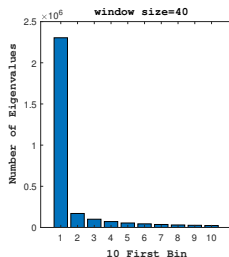
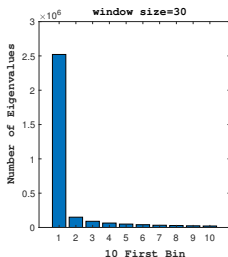


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- **Histogram** of eigenvalues over different window sizes.



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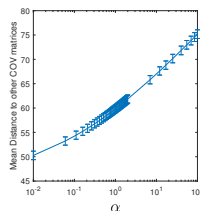
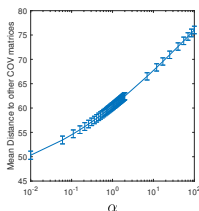
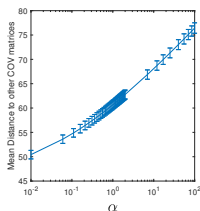
Step1



- Adding Identity matrix ($\tilde{\Sigma}_i + \alpha I$).
- What is the **impact** of α ?

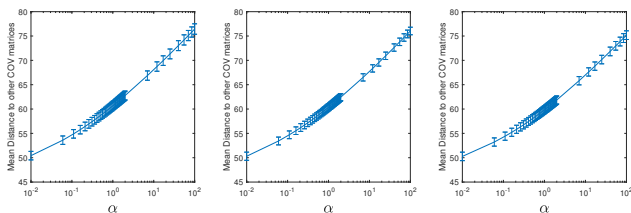


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- Analysis of barycenter based on Alpha





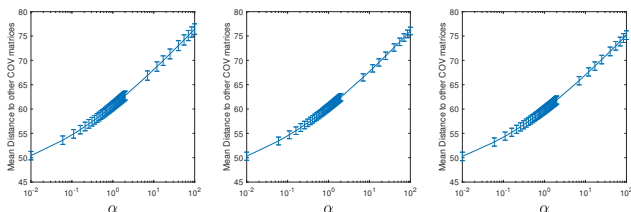
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- α has quite similar effect on all trials.
- Different trials have similar radios therefore we can choose an α for computing all barycenters.
- We **do not need to adapt** α to each trial.



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 - We **do not need to adapt** α to each trial.
- ✓ **Can we do better?**



- Computing Non-stationarity

$$\Gamma(i) = \mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) \quad (3)$$

Where $\mathcal{F}$ is Non-stationary measure.

- Possible choices for $\mathcal{F}$
 - Frobenius Distance
 - Kullback-Leibler Distance
 - Riemannian Distance
 - Optimal Transport Distance



- Frobenius Distance

$$\mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) = \frac{\|\bar{\Sigma} - \tilde{\Sigma}_i\|_F}{\|\bar{\Sigma}\|_F} \stackrel{\bar{\Sigma} = V\Lambda V^T}{=} \frac{\|\Lambda - V^T \tilde{\Sigma}_i V\|_F}{\|\Lambda\|_F} \quad (4)$$

$$= \frac{\|\Lambda - \mathcal{P}_V(\tilde{\Sigma}_i)\|_F}{\|\Lambda\|_F} = \mathcal{F}(\Lambda, \mathcal{P}_V(\tilde{\Sigma}_i)) \quad (5)$$

- Riemannian Distance

$$\mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) = d_{\mathcal{R}}(\bar{\Sigma}, \tilde{\Sigma}_i) \quad (6)$$

$$\stackrel{\bar{\Sigma} = V\Lambda V^T}{=} d_{\mathcal{R}}(\Lambda, \mathcal{P}_V(\tilde{\Sigma}_i)) \quad (7)$$

$$= \mathcal{F}(\Lambda, \mathcal{P}_V(\tilde{\Sigma}_i)) \quad (8)$$

Where $\mathcal{P}_V(\cdot)$ is projection operator on eigenbasis of $\bar{\Sigma}$.

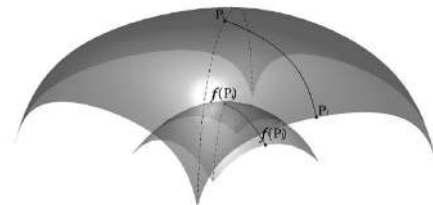
- How non-diagonal is $\mathcal{P}_V(\tilde{\Sigma}_i)$?

✓ **How to compare Λ and $\mathcal{P}_V(\tilde{\Sigma}_i)$?**

Results



- BSMML algorithm
 - Dimensionality Reduction
 - Minimizing the Riemannian geodesic distance loss between any pair of data points on the original manifold and sub-manifold of projected points.



- Computing Riemannian average of different classes
- Joint Diagonalization of them
- Finding a linear mapping to reduce the dimensionality
- Classification based on the sub-manifold.



Dadras Ali

Dear Dr Xie, Thank you for your completely message. I perfectly understand your si...

Thu 11/1, 11:51 PM



谢小峰 <xfxie@outlook.com>

Thu 11/1, 2:33 PM

Dadras Ali

Reply all

Dear Dadras Ali

I remember that you say you are working on the preprocess method research, so you can totally use your preprocess data on the BSMML to evaluta your preprocess method. Indeed, the preprocess step is very very important for the BCI system. Many BCI systems of our Lab are using this preprocess method. And some of BCI products will sell to the company/hospital. It is not my own work. That is why I can not offer more preprocess details with you. Hope you can understand me! Many Thanks!

Best regards,
Xiaofeng Xie



Dadras Ali

Wed 10/31, 10:59 PM





谢小峰 <xfxie@outlook.com>
Sun 11/4, 3:11 PM
Dadras Ali

Dea Ali Dadras,
Our preprocessed EEG data will be released after finishing our extension works.
Thanks for your email.

Best regards,
Xiaofeng Xie



Dadras Ali
Thu 11/1, 11:51 PM

Dear Dr Xie,
Thank you for your completely message.
I perfectly understand your situation.
But as you know I need to reproduce state of the art papers (like your paper) results to compare the improvement of results with our preprocessing method.
And I also understand you can't share other's work with me



- Accuracy for subject 1

	TSSM+LDA	TSSM+SVM	MDSM	TS+LDA	MDRM	NonStat+TS+LDA	NonStat+MDRM	CSP+LDA	CSP+SVM
S01	81.8	80	82.2	80.5	77.8	82.99	80.56	78.3	76.3



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S01	81.8	80	82.2	80.5	77.8	82.99	80.56	78.3	76.3

- Filter Banks
- EEG guy (preprocessed data)
- Dynamic Covariance Estimation

Future Works



- Wasserstein Stationary Subspace Analysis ¹
- There are two kind of source for EEGs.

$$E(t) = [A_s \ A_n] \begin{pmatrix} S_s(t) \\ S_n(t) \end{pmatrix} \quad (9)$$

- Minimization of sum of Wasserstein distances to find non-stationary sub-space.
- Finding the best orthogonal linear transform to project original signal to stationary sub-space.

¹Kaltenstadler, Stephan, et al. "Wasserstein Stationary Subspace Analysis." IEEE Journal of Selected Topics in Signal Processing (2018).



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- Finding the best orthogonal linear transform to project original signal to stationary sub-space.
- ✓ **Can we find a transform matrix insted of our one dimentional non-stationary measure $\Gamma(i)$?**

¹Kaltenstadler, Stephan, et al. "Wasserstein Stationary Subspace Analysis." IEEE Journal of Selected Topics in Signal Processing (2018).



- We can modify Riemannian metric.

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- How about separating $\mathcal{P}_V(\tilde{\Sigma}_i)$ into diagonal and non-diagonal matrices.

$$\mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) \simeq \underbrace{\mathcal{F}_1(\Lambda, \mathcal{P}_V^{diag}(\tilde{\Sigma}_i))}_{\text{Leakage of energy through eigenvalues of barycenter}} + \underbrace{\mathcal{F}_2(\Lambda, \mathcal{P}_V^{non-diag}(\tilde{\Sigma}_i))}_{\text{Leakage of energy from eigenvalues of barycenter}} \quad (11)$$



- Lets forget about EEGs and work directly with sample covariances.
- Is Barycenter the best representation of this set?
- Can we find a better representation for set of sample covariance matrices?

$$C_k = \sum_i \mathcal{G} \left(\mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) \right) \tilde{\Sigma}_i \quad , \quad \sum_i \mathcal{G} \left(\mathcal{F}(\bar{\Sigma}, \tilde{\Sigma}_i) \right) = 1 \quad (12)$$

Where C_k is the best estimated covariance matrix of k^{th} trial.

Thanks for your attention.